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Designing visualizations of temporal relations for children: action research meets HCI

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Abstract The diffusion of ICT products in everyday life is bringing into the realm of technological artifacts new users and their demands, which translate into novel design/research issues, like in the case of children-oriented Technology Enhanced Learning (TEL) systems. In this paper we report our experience within the framework of the EU project TERENCE, which developed an adaptive learning system supporting 7–11 years old children characterized by "poor text comprehension", a cognitive disability related to reading activities. In particular we discuss how our design approach allowed us to overcome the inadequacy of existing consolidated techniques for visualization of temporal relations and data gathering, which turned out not to comply with constraints typical of children-oriented TEL systems. We show how the definition of a novel children-oriented data gathering technique allowed us to carry out a study on the children mental model of time and, based on it, to design a children-oriented integrated "read-n-play" visual environment able to force the acquisition of the capability of reasoning about temporal events within a story. Since the aim of the stimulation is to drive children towards a mature mental model of time, the challenge was to overcome the drawbacks of adult-oriented techniques by building on concepts already possessed by children, while at the same time retaining some basic features of adult-oriented temporal visualization to favor the transition.

Keywords Children-oriented design · Temporal relations · Visualization · Action research · Games

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1 Introduction

The success of technological products highly depends on a good contextualization and on a clear understanding of needs, cultural constraints, and behaviors of the persons using them. A rigorous design approach requires that solutions be iteratively developed using state-of-the-art knowledge from existing theories and successful design practices, with reference to both the characteristics of the desired product (e.g., interaction and visual aspects) and the design process itself (e.g., methods and theories). One must also observe that, on the other hand, the growing diffusion of innovative digital artifacts in everyday life is extending the population of users in number and typology, bringing into the realm of ICT products people traditionally not served by technology, like, e.g., elderly people, children, people with special needs, technology-illiterate people. In these novel scenarios, not only designers and researchers have to rely on a lighter and sometimes less structured body of specific knowledge, but even consolidated techniques and methods may turn out to be less efficient, sometimes just when and where they become even more crucial than in more traditional application domains.

Designing for children is one such case in which designers experience the failure of principles, techniques, and solutions conceived for adult users. A recent paper on Child–Computer Interaction, the area of scientific investigation concerning the phenomena surrounding the interaction between children and computational and communication technologies, clearly indicates that "*... there is an urgent and present need for research in the design of interactive products for children, related methodology, as well as a scientific account of the interaction between children and technology*" [55]. In order to enrich the body of knowledge related to children-oriented design, it is underlined the necessity of empirically grounded guidelines taking into account the actual context in which children-oriented technology is used and able to capture the unstructured ways in which children use IT products [36]. Design and research activities should be founded on approaches that simultaneously assist in practical problem solving and expand scientific knowledge by generating and validating theories; in this way both relevance and rigor to the project at hand would be guaranteed. A beneficial side effect, one would achieve rapid dissemination of findings, necessary since guidelines shortly become obsolete because children in one decade tend to have more experience with ICT artifacts than children from the previous decade [36].

We experienced issues of this kind within the context of a Technology Enhanced Learning (TEL) project, aimed at developing TERENCE, an Adaptive Learning System (ALS) able to support 7–11 years old children characterized by "poor text comprehension", a particular cognitive disability related to reading activities (to give an idea of its diffusion, studies show that in UK about 10 % of children in the 7–11 age range are assessed as poor comprehenders [11]). While, traditionally, the psycho-pedagogical intervention on poor comprehenders is carried out by primary school educators by means of paper-based learning material, TEL systems opens new possibility with respect to the support to both teachers and learners. Since the few existing ALSs promoting reading interventions were all developed for old children or adults (see e.g., [9], and systems like QuizGuide, PROLEARN, PUPPYIR, ROLE[1]), TERENCE aimed at filling this gap by focusing on primary-school children.

[1] QuizGuide: http://www.pitt.edu/~taler/QuizGuide.html; PROLEARN: http://www.prolearn-project.org/index.html; PUPPYIR: http://www.puppyir.eu; ROLE: http://www.role-project.eu.

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1.1 Developing a technology enhanced learning system for "poor comprehenders"

Reading is recognized as a complex cognitive activity that transforms print to meaning through a negotiation between the text and its reader, as an activity of problem solving [60], which, according to [31], can be viewed as a multidimensional process including *decoding* and *comprehension*. Although correlated, these two skills depend on different cognitive and linguistic skills [49] and thus researchers classify poor readers in *poor decoders* and *poor comprehenders,* with distinct cognitive and linguistic profiles [30]: while poor decoders, often defined as dyslexics, have difficulties with fluent reading yet manage to comprehend what read reasonably well [65], poor comprehenders read words and sentences accurately, fluently, and at age-appropriate levels, but have serious difficulty understanding what they have read [46].

Stories have been recognized as a first class tool for psycho-pedagogical stimulation plans oriented to poor comprehenders [64]: in particular, since studies experimentally prove that poor comprehenders fail to master (1) the coherent use of cohesive markers such as temporal connectives (e.g., *before*, *after*, and *while*), (2) inference-making from different or distant parts of a text, and (3) their coherent integration [12], educational interventions aimed at improving text comprehension stimulate learners to reason about stories designed so to include appropriately interspersed temporal connectives through which children construct relations about story's events (e.g., [10]). Experiments also show that enriching inference-making questions with visual aids is pedagogically effective in fostering deep comprehension of stories, coherently with the *dual-coding theory*, which states that verbal and pictorial information, processed separately along distinct channels in the human mind, re-enforce each other when adequately paired [51].

Starting from such considerations and based on consolidated results about games as a means for fostering deep learning (e.g., [38, 49]), the TERENCE learning approach is grounded on the ideas of "learning through gaming" and "training via iterations": children play and enjoy two/three sessions per week collectively run in a dedicated classroom under the vigilance of a teacher, while each learner has his/her own individual experience with the system. The *stimulation plan*, designed by the psycho-pedagogical experts involved in the project, is grounded on two basic principles: (P1) learners have to be presented with material at an appropriate level of difficulty but nonetheless challenging; (P2) progresses in text comprehension are to be achieved by associating reading and playing. The primary guide for system adaptation mechanisms is the *reading comprehension* (RC) skill, introduced by TERENCE experts and measured on a scale with four levels taking into account comprehension of lexicon, grammatical skills, global coherence and local cohesion skills. RC skills are also the guiding concept for the *graded text simplification* approach introduced by TERENCE: to obey to principle P1, TERENCE stories are versioned and presented in four difficulty levels; to obey to principle P2, inference-making about stories is achieved through question-answer smart games classified into different levels of difficulty to stimulate text comprehension according to increasingly demanding tasks (in particular *temporal games* focus on temporal relations BEFORE-AFTER, BEFORE-WHILE, WHILE AFTER, and BEFORE-WHILE-AFTER).

While for extensive details on the graded text simplification and on the game-based TERENCE stimulation plan we refer to [3, 20], here we focus on the issues we had to face to design children-oriented visualizations of temporal relations that effectively support the cognitive tasks of poor comprehenders in the learning process, discussing the inadequacies of existing approaches and the design paths we followed to define new ones.

From what said so far, it is clear that the design of an educational system has to provide curricular materials (e.g., stories) and "assessment" strategies, not to be confused with

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"evaluation": while evaluation refers to the system, assessment refers to learners, judging their performances in terms of the psycho-pedagogical outcomes [8]. In an adaptive learning system assessment is a core concept of the system itself, which allows designers to evaluate the psycho-pedagogical effectiveness of the system strategy, i.e., the educational value of the system. This has to be done within the context of an iterative design, conducted as a "test and make changes" process, according to *formative evaluation*, which is "user testing with the goal of learning about the design to improve its next iteration" [48]. After formative evaluation and iterative design are complete, a final *summative evaluation* serves to document the effectiveness of the design and justify its use by learners and teachers [8].

An important aspect to consider is the age of target users of the system, which greatly affects the design approach to be used. Nowadays, one of the consolidated theory in the CCI research field is the Druin children's roles in a project, based upon a review of the literature and her own lab's research experiences in designing children-oriented technologies [26]. Druin has defined four distinct roles that children can play in the design process: (i) *user*: children contribute to the research and development process by using technology; (ii) *tester*: children test prototypes of technology that have not been released to the world; (iii) *informant*: children play a part in the design process at various stages, based on when researchers believe children can inform the design process; (iv) *design partner*: children are considered to be equal stakeholders in the design of new technologies throughout the entire experience. As to the fourth case, Druin proposes approaches that allow working with children as partners according to a co-design vision, but such approaches are laboratory-based and overtly not applicable in schools [27, 36]. There is some example of co-design at school (e.g., [63] explores the application of co-design with children in 7–9 age range), but, when situated within school activities, co-design shows limits when applied with many learners and organizational constraints (for example, as we will discuss later on, schools involved in the TERENCE project imposed that all children of a class had to be involved at the same time and that data gathering sessions had not to be longer than one hour).

It hence appears that in the case of the design of a TEL system meant for in-class activities it would be appropriate to involve children as *informant* throughout the design. Though the necessity of formative and summative evaluation within the framework of an iterative design makes User Centered Design (UCD) a good candidate design strategy, the generality of the UCD approach would not push designers to focus on the specific requirements of TEL systems, which are, among others:

R1. The necessity of designing both a psycho-pedagogical stimulation plan and the system modules realizing it in a context of mutual dependency;
R2. The definition of a psycho-pedagogical assessment strategy;
R3. The necessity to conduct large-scale studies within users' inquiry, assessment, and evaluation activities, to ensure statistical significance of results.

Furthermore, from the specific needs of psycho-pedagogical intervention on 7–11 years old poor comprehenders, there emerge, among others:

R4. The necessity of defining suitable children-oriented visual representations for temporal structures of stories.

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Coping with requirements *R1* and *R2* required us to specify the classical UCD schema (see Fig. 1) so to (1) explicitly address distinguishing activities/objectives of the design of educational software (see also [8]), (2) underline roles and responsibilities of different sub-teams of the multidisciplinary team, and (3) consistently coordinate design steps (e.g., just to mention a couple of cases, the psycho-pedagogical stimulation plan is designed only after the selection of the learning approach and its expert-based evaluation has to preceed – for ethical issues – the moment in which the system is evaluated with children).

Meeting requirements *R3* and *R4* was one of the TERENCE design challenges because of (1) the inadequacy of existing time-oriented visualization techniques to meet the demands of children in general and of poor comprehenders in particular, and (2) the inadequacy of existing adult- and children-oriented data gathering techniques to meet project ethical and organizational constraints, as discussed in the following two subsections.

1.2 Inadequacy of existing visualization techniques

Defining suitable visual representations for temporal structures of stories needs the integration of theoretical and methodological work both from traditional areas devoted to temporal representation (logic, reasoning, and databases) and from the information visualization research field [1]. First proposals for visualization techniques came from [33, 35], based on seminal Allen's work on temporal logic [2], (see Fig. 2). Stemming from these results, other researchers proposed visual metaphors and tools surveyed e.g., in [1, 13, 16]. More specifically, [13] presented results classified according to the four design dimensions of temporal structure, order, representation, and history, [16] classified literature proposals according to time granularity and capability of expressing disjunctive relations between pairs of intervals or points, while [1] uses time, data and representation dimensions.

Timeline is the most used technique for interacting with time-linear visual information (where a timeline is a graphical or textual display of events in chronological order [40]),

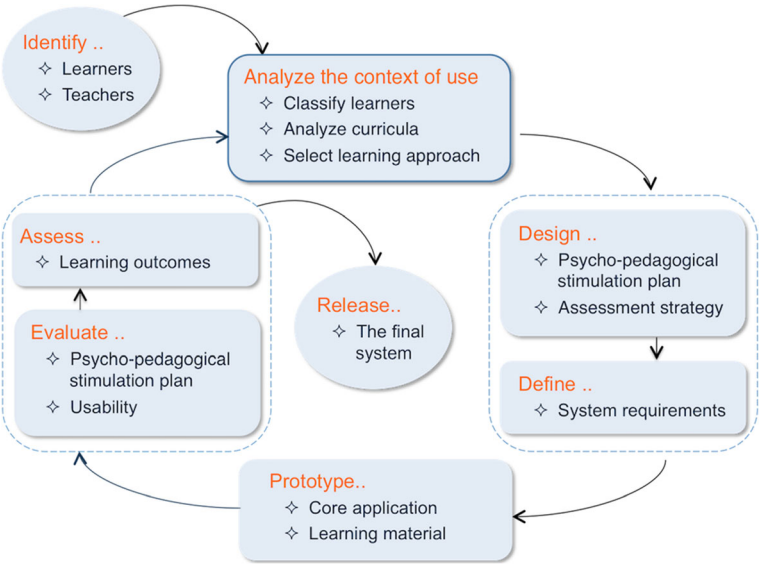

Fig. 1 TEL-oriented UCD schema

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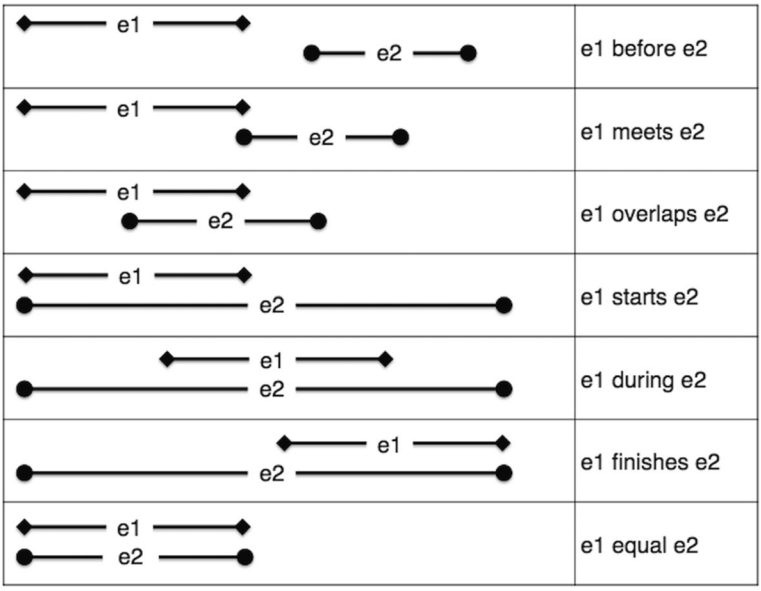

Fig. 2 Allen's temporal algebra

adopted in a number of interactive systems based on 2D or 3D visualizations. While for extensive discussions we refer to the aforementioned surveys, we want here to mention just a few examples of timeline-based proposals to give an idea of the complexity of such systems and being able, later on in this section, to assess existing techniques against requirements of children-oriented systems. In 2D visualizations the time dimension is typically represented "à la" Allen, generally equipped with scrollbars (needed when intervals of interest do not fit on the screen) and sliders labeled with discrete periods at specific time granularities (e.g., in Lifelines [52], time axis may be selected at several levels of detail, allowing the view of events at different time granularities - year, month, day).

Some systems integrates 2D and 3D visualizations, like the Perspective Wall [43], where data (contextual information) outside the center focus area (detailed information) are reduced in size, and represented in two perspective areas, or Time Tubes [15] composed of one or more Disk Trees, which represent the hyperlink structure of a Web site, aimed at organizing and visualizing the evolution of Web sites over specific time periods (see e.g., http://www.parc.xerox.com/istl/projects/images/). An approach integrating timeline and calendar views is TimeScape [57], where the user may access past and future desktop states of personal information in a desktop environment, and zooming mechanism is used for changing the time granularity of the timeline.

Considering the entirety of relations within a whole story (which is our final objective) makes the situation even more complex, since it means moving from linear representations to graph and network representations (a proposal bridging the two cases is in [16], which introduces three alternative visual metaphors that can scale up to the relation visualization in a network with more than two intervals; these metaphors, based on concrete objects and phenomena – like elastic bands, springs and paint strips – can also render networks of more than two events, and their relations). It has to be observed that in this case *temporal annotation* of stories is an indispensable yet complex task posing specific challenges deriving, among

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others, from high density of temporal relations and distances within the text of involved objects (e.g., events) [66]. Several tools have been proposed to deal with the complexity of temporal annotation and the consequent low markup speed, hard-to-avoid inconsistencies, and low inter-annotator agreement, like, e.g., PatternFinder tool [28], BAT,[2] TANGO,[3] T-BOX [66] and TARSQI,[4] and TATOT [21], most of them based on Graphviz,[5] an open source graph visualization software. Anyhow, all these tools are oriented to an experienced story creator rather than to the story reader.

In summary, while adequate for adult professionals such as computational linguists, information engineers, or story creators, existing visual representations fail to directly support the requirements of young learners, for a number of reasons:

- *High degree of abstraction*. Almost all proposals are based on abstract elements, and even metaphors based on real objects, such as paint strips, turns out to be not realistic enough, while recent studies prove that children appreciate realistic illustrations [14].
- *High degree of determinacy*. In almost all proposals, events are meant to represent time interval with known duration, and starting and ending points, while children-oriented story are characterized by time indeterminacy and vague temporal relationships ("Once upon a time …", "While Gino the Chick is telling stories to the wolf, Mammy Hen and the others animals go to the wolf's house") [64].
- *Fine granularity.* Based on Allen's work on temporal logic, all proposals aim at clearly differentiating among distinct configurations of non overlapping intervals (the first two cases in Fig. 2, i.e., relations *before* and *meets*), or overlapping intervals (the other cases in Fig. 2, i.e., relations *overlaps*, *starts*, *during*, *finishes*, and *equals*), arriving to a level of detail that turns out to be excessive and counterproductive for children novice in the mastering of the general meaning of BEFORE, AFTER and WHILE [44].
- *Lack of global vision.* None of existing proposal aims at relating events to the global context of a narration, while stories are a first class tool of a psycho-pedagogical stimulation plan [64].
- *Groundness on a pre-existing model.* Visual representation of temporal data and temporal connectives are generally based on (crowded) timelines and visualization of intervals. Anyhow, it has to be underlined the crucial difference between adult-oriented time representation and child-oriented time representation aimed at supporting text comprehension: in the first case we build on a pre-existent mental model of time and temporal connectives, while in the latter case we have to induce the construction of a mental model of time and temporal connectives. Relying on his/her mental model, an adult can grab temporal information from the exploration (and possibly the filtering) of timelines (often characterized by huge quantity of data) that would on the contrary overwhelm a child who is trying to acquire common sense time concepts. Furthermore this would be inconsistent with consolidated pedagogical approaches built on question-based games [64].

All this said, one of the primary tasks of TERENCE designers was hence the definition of a children-oriented integrated "read-n-play" visual environment able to force the acquisition of

[2] http://www.timeml.org/site/bat/
[3] http://www.timeml.org/site/tango/
[4] http://www.timeml.org/site/tarsqi/toolkit/
[5] http://www.graphviz.org/

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the capability of reasoning about temporal events within a story. Since the aim of the stimulation is to drive children towards a mature mental model of time, the challenge was to overcome the above discussed drawbacks by building on concepts already possessed by children while at the same time retaining some basic features of adult-oriented temporal visualization to favor the transition. An extensive study involving children of different age and different RC skills was hence necessary to understand the mental model of time of children, how it is related to age and RC skill, and how it evolves with age. This study had to be the mandatory starting point of the iterative design process of TERENCE.

1.3 Inadequacy of data gathering methods

We converged to the release of the final TERENCE system in three UCD iterations, deploying prototypes subject to formative evaluation and the final system subject to summative evaluation, to create a solution usable and pedagogically effective. The study on children took place within the context-of-use analysis (the bordered block in Fig. 1) of the first UCD iteration that, overall, included a preparatory study and two rounds of field studies.

The *preparatory study*, aimed at understanding how children are assessed by psychologists as poor comprehenders, was carried out by brainstorming with about 30 domain experts [59]. A relevant outcome was the identification of topics to be investigated by field studies to be conducted in schools in UK and Italy according to indications from school deans and teachers: though researchers underlined the need of direct interaction with children within schools (since the system is intended to be integrated into regular school activities), school bodies raised organizational and ethical issues (e.g., they initially requested that data gathering would not interfere with in-class activities) thus forcing the project team to structure a *1st round of field studies* as a combination of customary expert-based and indirect user-based data gathering methods (e.g., diaries), according to approaches followed by the participating schools when involved in research projects. This first round, involving about 100 learners and 70 educators, allowed to define main requirements for the learning material but failed to obtain an adequate cognitive characterization of learners [59] due to limits of the adopted approach: first of all, since teachers administered diaries to children as homework to be done with family members, results were strongly adult-biased; furthermore, only a small quote (around 30 %) of the administered diaries were actually returned, moreover mostly poorly filled in. Summarizing, the indirect approach failed to meet desired requirements of genuineness, dependability, and statistical significance of gathered data.

A *2nd round of field studies,* necessary to enrich the results, had to be based on two principles: (1) some kind of direct interaction with learners was necessary to gather high quality data, and (2) a large-scale study was mandatory to ensure statistical significance and hence pedagogical effectiveness of the system. At this point schools agreed on the fact that data gathering activities could be conducted during regular school activities, but required that all children of a class had to be involved at the same time and that the duration of data gathering sessions had to be no longer than one hour. The overall requirement was hence to gather *high quantity of high quality data from many users in short sessions with medium size users' groups*. This requirement was assessed against consolidated adult-oriented and children-oriented data gathering techniques to detect the approach that could best fit the organizational situation (Table 1 reports a synoptic view of this analysis).

As for children-oriented data gathering techniques, in the last decade design theories evolved, adopted and/or adapted for work with children (see, e.g., [26, 36, 47] for in-depth

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Table 1 Assessment of existing techniques against project constraints

Adults/Children	Techniques	Pro	Cons
Adult-oriented and adaptable to children	Survey	High quantity of data statistically analyzable	Low data quality: many sources of bias
	Interviews	High data quality	Too long if many users
	Questionnaires	High quantity of data in relatively short sessions	Low data quality
	User observation and field studies	Viewing users in their real context	Interference with routine school activities
	Brainstorming	High data quality	Low data quantity
Children-oriented	User Interface Drawing	High quantity of data in relatively short sessions	Low data quality
	Idea Maps	High quantity of data	Low data quality
	Kid Reporters/Networking news	Inspiring social settings, high quantity of data	Interference with routine school activities, low quality (no specific user requirements)
	Mission from Mars	Playful and motivating framework, and high data quality	Interference with routine school activities and age groups (>= 10 years old)

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discussions). Differently from the more traditional techniques reported in the first group of Table 1, generally used with adults and adaptable to children, the techniques in the second group have been proposed in the context of projects dedicated to children in which they were involved as design partners. These techniques were used for directly questioning specific user data/requirements differently from others (e.g., Photo Diaries [50] and KidsReporter [7]) clearly designed and dedicated to children but more focused on involving children in the design activities then gathering data from them.

Thus, unfortunately, as summarized in Table 1, none of the existing approaches fit with all the constraints coming from TERENCE ethical and organizational requirements, and hence the need emerged for conceiving (and validating) an appropriate novel data gathering approach and applying it to the immediate problem situation.

1.4 Structure of the paper

The remainder of this paper discusses how we faced the so far analyzed inadequacies of existing approaches. While in [23] we discussed the design path the led us to the definition of a novel children-oriented game-based data gathering technique, in this paper we are interested in moving some steps forward and shifting the focus towards:

- How the technique was applied within TERENCE;
- The results we got about the children mental model of time;
- How these results translated into requirements for the “read-n-play” visual environment;
- How such requirements translated in turn into the TERENCE visual environment (some preliminary results on the visual interface appeared in [24]).

More specifically, Section 2 sketches the Action Research (AR) process that allowed us to address the multifaceted objective of generating, validating, and applying a new children-oriented data gathering technique, while, at the same time, translating the experience into scholarly knowledge related to design and evaluation of children-oriented applications. Section 3 brings into focus the major results of the AR process, namely, (1) the data gathering technique, providing guidelines in terms of the overall structure of the method and the design of individual games, and (2) its instantiation within the TERENCE project within the analysis of the context of use, sketching the overall set of games designed in association to the topics to be investigated, focusing in particular on games associated to temporal connective mastering, and discussing the results of the analysis of the game results. Section 4 shows how these results on the children mental model of time translated into the TERENCE visual smart games, whose usability and psycho-pedagogical effectiveness is analyzed in Section 5. Finally, in Section 6, conclusions are drawn.

2 Overcoming the inadequacy of existing data gathering techniques

Our first challenge was to address the multifaceted objective of (1) generating and validating a new children-oriented data gathering technique, (2) carrying out the analysis of the context of use coherently with constraints coming from the project (scheduling and budget/resources limitations) and from actors involved in the project (schools and experts), and (3) translating the experience into scholarly knowledge.

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Action Research (AR) allowed to satisfy this need: initially used in the social and medical science and more recently considered by the Information Systems (IS) field, AR is based on a juxtaposition of practice (action) and theory (research) and is committed to the production of scholarly knowledge through the seeking of solutions/improvements to "real-life" practical problem situations [4, 45] carried out collaboratively by researchers and an organizational "client": the "laboratory" of AR is the real world, thus avoiding the potential separation of research and practice. Since AR is originally based under the founding principle that complex social processes can be studied best by introducing changes into these processes and observing the effects of these changes, Baskerville observes that one area of importance in the ideal domain of AR is new or changed systems development methodologies [4]: studying new or changed methodologies indeed implicitly involves to introduce such changes, making AR appropriate for the novel design/research challenges raised by new users brought into the market by new ICT applications.

2.1 A brief introduction to action research

The origins of AR can be traced to the work of Lewin [41, 42] and of researchers at the Tavistock Clinic [62], which inspired a vast stream of work (see, e.g., [4–6, 18, 45] for discussions and surveys) all sharing the common notion of some kind of cyclical process repeated until a satisfactory outcome is achieved. The seminal paper of Susman and Evered [61] strongly influenced the field: according to their view, a client-system infrastructure (the *researcher-client agreement*) has to be established and then an iterative process based on a five phases Cyclical Process Model is carried out (Fig. 3a): in the *diagnosis* phase, primary problems causing the desire for change are identified and the theoretical framework that will guide the process is developed; in the *action planning* phase, the actions that should solve the organizational problems are specifies, based on the theoretical framework; in the *action taking* phase, planned actions are implemented; in the *evaluation* phase, the team determines whether the theoretical effects of the action were realized and whether they produced desired results; in the *learning* phase, the knowledge gained throughout the entire process is summarized and formalized, with reference to the immediate problem situation and the scientific community.

Two notable contributions aim at facing criticisms about lack of rigor and ambiguous resemblance with consulting (from which AR anyhow differs, being committed to the production of scholarly knowledge [4]): in [18] authos develops a rich and structured set of interdependent

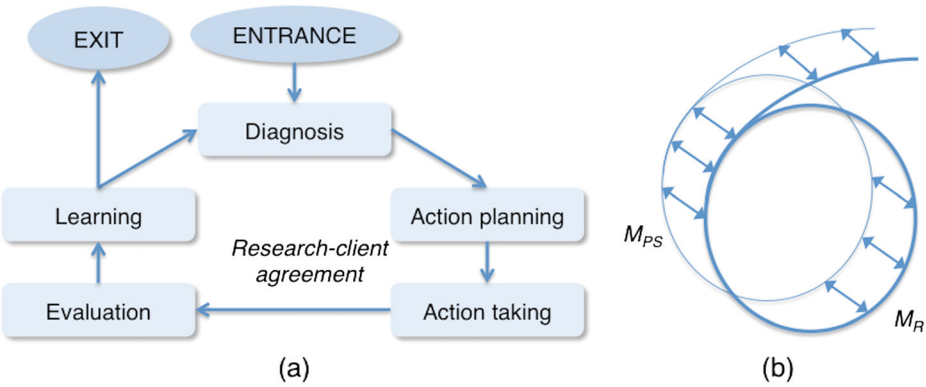

Fig. 3 Two models for the AR process

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Table 2 Elements of an AR intervention according to [45]

Elements of a tandem-based Action Research intervention	
A	A real world problem situation
P	A real-world example of A allowing the researcher to investigate A
F	A theoretical premise declared by the researcher prior to any intervention in A
M_R	The research method (i.e., Action Research)
M_{PS}	The method (M) which is employed to guide the problem solving (PS) intervention

principles and criteria that researchers and reviewers can use to ensure and to assess the rigor and relevance of Canonical Action Research (CAR), respectively, while the study in [45] argues that the ambiguity issue is related to the way in which AR is conceptualized and proposes an extended model including two cycles running in tandem (Fig. 3b), focused on distinct aspects: one cycle addresses the client's problem solving interest while the other one explicitly addresses the researcher's scholarly interest. As a consequence, two methods are being used, one (M_R) being Action Research itself, used to investigate on a real world problem situation (A), and the other (M_{PS}) being the one adopted for the problem solving of a real-world example of A (P); to comply with rigor requirements, a theoretical framework for M_{PS} is selected prior to the intervention (Table 2).

AR has actually recently received attention from Human-Computer Interaction (HCI) researchers, though the focus has been put more on the evaluation of new technology in novel contexts than on conception/evaluation of new design and development methods (see, e.g., [25, 32, 34, 39, 58]). A few examples of AR-based approaches can be found also in children-oriented projects [37].

2.2 Our action research project: a summary

We based our project on the CAR approach of [18] blending it with the tandem view of [45]; as to individual cycles of the tandem, we followed the Susman and Evered Cyclical Process Model [61]. Furthermore, since theoretical and empirical studies show that learners are more motivated to participate in school-class activities if they are shaped like games (e.g., [38, 53]), we decided to base the field study on games. Under this theoretical premise, in our case the elements of the AR intervention of Table 2 are instantiated as in Table 3.

As observed in the Introduction, while in [23] we focused our attention on methodological aspects and on the process that led us to the definition of the data gathering technique, we are here more interested in discussing the activities that took place on the problem solving side to

Table 3 Elements of our Action Research intervention

Elements of the TERENCE tandem-based Action Research intervention	
A	Issues and challenges in effectively gathering dependable children's data on a large-scale basis within school organization
P	Defining the user classification and the personas for the TERENCE project
F	Frameworks from consolidated data gathering techniques can be effectively blended with existing game frameworks
M_R	Action research
M_{PS}	The new untested evolving technique

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show how their results were used in successive phases of the project to realize the children-oriented visual learning environment. So, in this Section, we summarize the Researcher-Client Agreement and provide an overview of the Cyclical Process, while leaving to the next section an in-depth analysis of the results of individual stages of the CAR process (we refer the interested reader to [23] for a detailed discussion on our CAR project).

2.2.1 On the researcher-client agreement

The Researcher-Client Agreement (RCA) represented the grounding foundation guiding the establishment of focus, boundaries, and objectives of the project, defining roles and responsibilities of the participants, selecting data collection and analysis methods as well as measures to evaluate the results. Though for the sake of a neat presentation these aspects are all discussed here as pre-project agreements, we have to observe that in practice they are the result of continuous reflections and refinements throughout the entire CAR project.

The *working team* comprised two panels: a *school panel*, which was actually the client of the CAR, included representatives from the schools involved in TERENCE (for each school its dean and one teacher per participating class); (2) a *panel of experts* from the technical TERENCE team included two action researchers and people with background in data gathering, game frameworks, and psychology. It was agreed that the *focus of the CAR project* had to coincide with the context of use analysis phase of the TERENCE project and that, considering related literature and teachers' experience, the novel technique should have recourse to games as the primary tool for gathering data directly from children. Teachers stated ethical principles (*Eth1* and *Eth2*) and organizational constraints (*Org1* and *Org2*) to be satisfied:

Eth1 It would be unfair not to involve all children of a class at the same time (children might perceive unjustified differences in treatment);
Eth2 It would be unethical to make children invest time in activities that are not a value for them;

Org1 Activities has to be structured in sessions not longer than one hour;
Org2 Project activities have to be integrated into regular school activities.

To define *client commitment, roles and responsibilities,* the client and researchers signed protocol agreements aimed at regulating activities with children (among others, assuring school commitment, defining rules/roles for in-class activities, and relieving schools from any cost). Furthermore there was a word-of-mouth agreement related to the participation of teachers and researchers into all stages of the CAR process, with different roles: as to the design, researchers had to guide the overall process while the clients had to assess design choices; as to the implementation, it was requested by the client that researchers be active parts in game-based activities conducted with children, since, according to the experts' opinion, it would stimulate attention from children (this let researchers be close to the action while maintaining a clear separation in roles).

Objectives of the immediate problem situation and of the CAR project were refined throughout the process. Starting from the general question "Can data gathering be conducted through games?" research questions can be summarized as:

Q1 CONSTRAINT COMPLIANCE – Can a game-based technique be designed so to be compatible with given ethical and organizational constraints?

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Q2 EFFICACY – Can game design be guided by the topics to be investigated and identified in previous phases of a project (preparatory studies)?
Q3 EFFICIENCY WRT ACTION – Are data gathering in-class activities playful (i.e., involving children and guaranteeing a correct level of attention during the sessions)?
Q4 EFFICIENCY WRT ANALYSIS – Are data gathered through appropriately designed games (see Q1 and Q2) characterized by quality (genuineness and dependability) and quantity adequate to be statistically significant (the main limitation of diaries)?

To being able to *evaluate the CAR results*, two streams of data gathering were conducted in parallel by two design teams, one in UK and one in Italy respectively based on customary observational methods, and on the AR approach. The two design teams were characterized by comparable experience in UCD in general and in data gathering in particular, and worked under the same TERENCE time constraints, so to be able to compare the achieved results of the two teams. It was also agreed to be the comparison on data dependability, richness and specificity in users' class description, and ease in persona derivation, in order to provide a qualitative assessment of CAR results. Furthermore, teachers and deans were involved in intermediate and final brainstorming and semi-structured interviews. Results achieved by the CAR project were also disseminated and assessed outside the consortium via workshops dedicated to HCI researchers and other stakeholders.

2.2.2 On the cyclical process

Overall, our cyclical process required three iterations: the first two iterations were focused on trial and assessment of the techniques (which hence implied two rounds of field studies), while the third one included analysis of gathered data, users' classification and personas definition. Figure 4 provides an overview of the CAR process by illustrating the relationships between the two cycles in term of time scheduling (what happens before/while what) and information exchanged (represented by dashed inter-cycle arrows). Some notable characteristics may be observed:

- *Diagnosis* of the organizational situation was the first to start on the problem solving side ($\text{DIAGNOSIS}_{\text{PS}}$), and informed activities of diagnosis on the research interest side ($\text{DIAGNOSIS}_{\text{R}}$), which singled out, surveyed and assessed relevant literature with reference to the immediate problem to solve (actually, the considerations about the inadequacy of the results of the first round of field study and of existing approaches discussed in the Introduction are – de facto – part of the $\text{DIAGNOSIS}_{\text{PS}}$ and the $\text{DIAGNOSIS}_{\text{R}}$ stages of the 1st iteration). Objectives and research questions were singled out/refined in the diagnosis stages.
- *Action planning on the research interest cycle* ($\text{ACTION-PLANNING}_{\text{R}}$) includes activities related to planning and designing the research project so to adequately address the research questions, validate hypothesis and in general guide the overall process. It was in this stage that, in the 1st iteration, it was hypothesized that at least two iterations had to involve intervention in schools (trial and assessment of the technique) and that at least one iteration had to be focused on the analysis of gathered data. Accordingly, schools were aggregated in two groups.
- *Action taking on the research interest cycle* ($\text{ACTION-TAKING}_{\text{R}}$) was focused on the design/refinement of the data gathering technique ("method design" block in Fig. 4) and had necessarily to start before the game design within the action planning stage of the other

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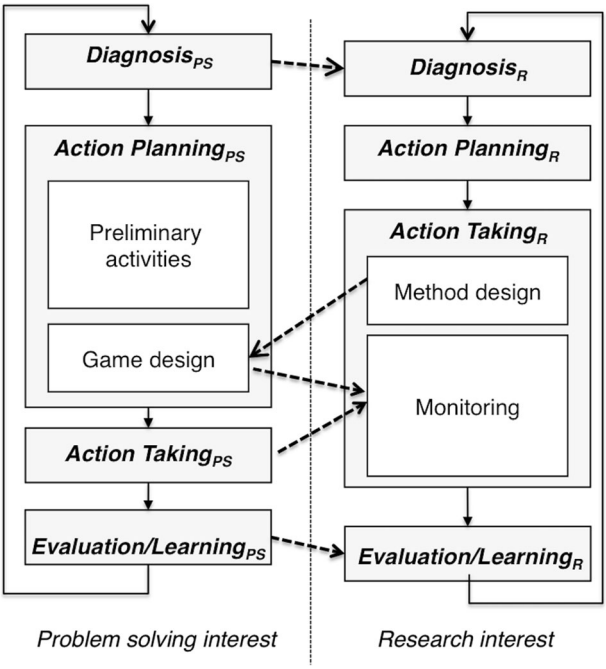

Fig. 4 Our tandem CAR cyclical process

cycle: only once the method (or an intermediate versions of it) was available, the team (client and researchers) could start to design the games, and, after that, implement the TERENCE data gathering. It has to be noticed that, while action planning and action taking on the problem side (ACTION-PLANNING$_{PS}$ and ACTION-TAKING$_{PS}$) were being carried out, some researchers were monitoring these activities (the "monitoring" block in Fig. 4).

- *Action planning on the problem solving interest cycle* (ACTION-PLANNING$_{PS}$) has to specify organizational actions that should relieve or improve the identified organizational problem. In our case, after preliminary activities independent from the availability of the data gathering technique (e.g., interviews with stakeholders), it was primarily focused on designing/refining games and game administration protocols (the "game design" block in Fig. 4), but for the 3rd iteration when the focus shifted from data gathering to data analysis and activities of this stage mainly concerned setting up game result analysis.
- *Action taking on the problem solving interest cycle* (ACTION-TAKING$_{PS}$) has to implement the planned actions, with the client and the researchers collaborating in the active intervention. In our case, it was primarily focused on the administrations of games in selected schools, but for the 3rd iteration when the focus shifted from data gathering to data analysis and the team analyzed data, derived classes of users, and designed the associated personas.
- As to *evaluation and learning* stages (EVALUATION/LEARNING$_{R}$ and EVALUATION/LEARNING$_{PS}$), the team (1) evaluated the effect of actions on the problem and assessed them against research questions, so to determine if the results were satisfactory, and then exit, or, otherwise, to single out unsolved issues to be addressed in the next iteration, where previous results were amended, and (2) reflected on gained knowledge both on the immediate problem level and on the generalization of its solution according to research questions.

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A particular attention has to be given to the evaluation activities of the 3rd iteration, when the TERENCE consortium discussed and compared the results achieved by the Italy and UK teams through restricted and plenary meetings. A first appreciation of the work of the Italian team was related to the number of children that the characteristics of the adopted approaches allowed to involve: the two rounds of field studies in Italy (1st and 2nd CAR iterations) involved about 50 % more children than the single round of observational studies in UK *within the same period of time*. Furthermore the consortium agreed on the fact that users' classes produced by the AR process were richer and more specific than the ones produced by the UK observational study, and that personas were more easily derivable. Consequently, the CAR results were taken as a reference for the continuation of the project.

3 Focusing on main results of the AR-project

We want now give a closer look to results from the two sides of the CAR project, presenting the final outcomes of the AR cyclical process, referred both to the research interest side and to the problem solving interest side of the process. In particular, we will present:

- The main features of the novel data gathering technique, as defined in the research interest side during ACTION-TAKING$_R$ (1st, 2nd, and 3rd iteration were respectively focused on trial, refinement, and final definition of implementation instructions of the technique);
- The TERENCE investigator kit, as defined in the problem solving interest side during ACTION-PLANNING$_{PS}$ (1st and 2nd iteration);
- The administration of the games, as implemented in the problem solving interest side during ACTION-TAKING$_{PS}$ (1st and 2nd iteration);
- The analysis of game results, focusing on those aspects related to the children mental model of time, carried out during the ACTION-TAKING$_{PS}$, EVALUATION/LEARNING$_R$ and EVALUATION/LEARNING$_{PS}$ stages of the 3rd iteration (both sides of the process were involved).

3.1 The game-based children-oriented data gathering technique

As more traditional direct investigation techniques, our method is based on both a *direct observation* during game administration and a subsequent *indirect observation* on collected materials (e.g., game results, notes, and audio/video recording). Moreover, since we worked under the general assumption that the game framework has to be blended with frameworks from consolidated data gathering techniques (see Table 3), the method was structured as a sequence of the three customary stages of *planning*, *running,* and *reporting* (see Fig. 5). Notice that, to face with given constraints, the running stage is in turn structured as a number of one hour game sessions, to collect as many data as possible from a number of different classes; though theoretically independent, it may be useful (and actually necessary if conducted by the same investigators) that game sessions are not run simultaneously, as it will be clarified later on.

The running stage is designed as a sequence of customary phases (see Fig. 6): *nurturing* (investigators introduce themselves, explain the aim of the session, and establish a playful atmosphere), *motivation* (based on techniques from motivational theory [54]), *body*

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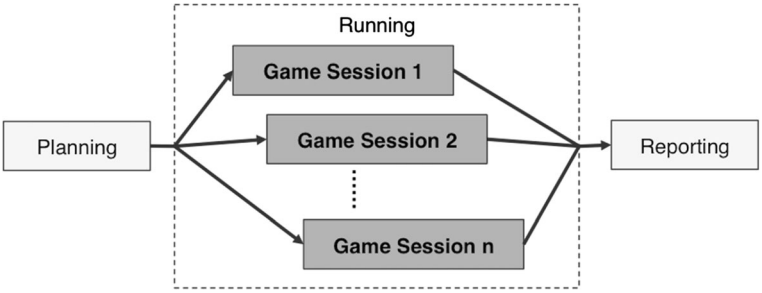

Fig. 5 The overall structure of the method

(investigators administer games and observe children), and *closing* (each child gets a reward, investigators reorder collected material and write down first impressions). The body, in turn, is designed as a cyclical iteration of four steps (one iteration corresponds to the administration of one game): *energizing* (investigators introduce goals, moves, and game rewards), *playing* (the actual game takes place), *rewarding* (investigators assign prizes to game winners), *reorganizing* (investigators dynamically adjust the session planning based on how activities are going).

To guarantee the uniformity of individual game sessions, the planning stage has to define a common "investigator kit", including games, game material, note templates, and a database to be populated by data gathered during the running stage. The kit must include a specific game for each specific topic singled out during preliminary studies and to be investigated. Properties of the overall set of games and of individual games were defined/refined during the entire CAR process and are described in Tables 4 and 5.

3.2 The TERENCE investigator kit

According to the method definition provided by ACTION-TAKING$_R$, the team designed six general games (to investigate on general aspects related to the relationships of children with games and technology) and two time-related games (to investigate on the children mental representation of BEFORE-AFTER and WHILE temporal connectives), all compliant with playfulness, child personal enrichment, and ethical issues, exploiting also preliminary pre-project results on the children mental model of time [17]. Games were checked and assessed them with schoolteachers (e.g., if a challenge was judged too difficult or too boring for a school class, it was revised according to teachers' feedback). These activities involved action researchers, experts in data gathering, one expert in game frameworks, one psychologist, and the teachers of all participating classes. Table 6 provides a synoptic view of the final version of the overall set of games as refined within ACTION-PLANNING$_{PS}$ of the 2nd CAR iteration.

As to the design of individual games, we exploited results of [54], where authors overview research findings about the correlations between the appeal of games and the psychological need satisfaction they provide, and propose a motivational model. In particular they show that, besides

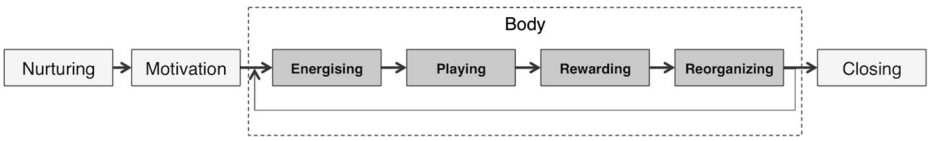

Fig. 6 The structure of a session

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Table 4 The description of the investigator kit. Properties I and II are outcomes of the 1st CAR iteration, while Properties III and IV are outcomes of the 2nd CAR iteration. The refinement was determined after the evaluation of the 1st iteration in which was observed that during game administration some children preferred not to participate in the game activities, some children lose interest during the game, and in several classes it was not possible to carry out all designed games within one hour

The investigator kit	
Components	
	Framework instantiations for individual games
	Game material
	Note templates
	Database schema
Properties	
I	The overall set of games has to include a specific game for each specific topic singled out during preliminary studies and to be investigated
II	Each individual game in the kit has to satisfy properties in the Annex (see Table 5)
III	The overall set of games has to include games with different cognitive load so that the body of a game session mirrors customary warm-up, peak, and relaxing phases
IV	Topics (and associated games) have to be prioritized according to their relevance to the project in order to be able to shape the individual game session on the fly at running time (to adjust the session depending on interrupts and other unpredictable events) while maintaining a warm-up/peak/relaxing structure and possibly sacrificing some games

the basic elements of *move* of the player and *outcomes* showing progresses, at least three factors determine engagement: *autonomy*, amounting to a sense of choice and psychological freedom (e.g., players may choose the level to play or the avatar), *competence*, realized by carefully balancing the game challenges to the players' skill, and *relatedness needs*, i.e., the sense of communion with others, attained by stimulating collaboration or competition. This motivational model was selected as grounded theory for the design of the individual games; based on it, a framework for game specification was created and games were defined specifying goals and moves and how autonomy, competence and relatedness needs were pursued. For each game, the team created game material, notes' templates, and the database to be populated by gathered data.

Table 5 Properties of individual games. Properties I to IV are outcome of the 1st CAR iteration, while Properties V to VII are outcome of the 2nd CAR iteration. The refinement was determined after the evaluation of the 1st iteration in which was observed that during game administration some children preferred not to participate in the game activities, some children lose interest during the game, and in several classes it was not possible to carry out all designed games within one hour

Annex: Properties of individual games	
I	Each individual game must cover all subtopics of the topic it is associated to
II	Playfulness, child personal enrichment, and compliance with ethical issues are mandatory characteristics of individual games
III	The estimated duration of a game session should not exceed 45/60 min (see *Org1*)
IV	Individual games should produce children-generated collectable results (e.g., drawings, conceptual maps)
V	Each game is characterized by a cognitive load (warm-up/peak/relaxing)
VI	Each game is characterized by an estimated duration (typically qualitative: short/ medium/long)
VII	Each game must include a rewarding mechanism, designed so to stimulate the production of genuine data from each child

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Table 6 The overall set of games designed for the TERENCE data gathering activities

Game_id	Cognitive load	Topic / subtopics	Type	Priority	Duration
W1	Warm-up	Extra-curricular activities (*what, where, when*)	Single	1	Short
W2	Warm-up	Time spent with parents (*what, where, when*)	Single	2	Medium
P1	Peak	BEFORE-AFTER relation (*spatial strategy/ies*)	Single	1	Short
P2	Peak	WHILE relation (*spatial strategy/ies*)	Single	1	Long
P3	Peak	Use of game consoles (*which, when, with whom*)	Group	2	Medium
P4	Peak	Use of technology (*which, where, what, when*)	Single	2	Short
R1	Relaxing	Preferences on game characters (*who, why, in which game, ...*)	Single	1	Short
R2	Relaxing	Homework activities (*when, how, with whom, ...*)	Group	2	Medium

Tables 7 and 8 summarize the features of the games designed to investigate on the children perception of the BEFORE-AFTER and the WHILE temporal connectives. It has to be noted that games produce collectable results that can be later analyzed by the team during the indirect observation phase of the technique.

Table 7 The BEFORE-AFTER game

BEFORE-AFTER game	
Investigation objective	
To gather information about the mental representation of the BEFORE-AFTER relation	
Game material	
Kits of transparencies depicting story characters, A4 paper sheets	
Game description	
Preliminaries	The investigator read a story and provides each child with a kit of transparencies and an A4 paper
Goal	The goal of the challenge is to provide a composition of transparencies, positioned on a sheet of paper, after the investigator reads a sentence containing a BEFORE-AFTER relation
Moves	Each learner has to choose transparencies from a kit s/he has received and to position them on an A4 paper
Engagement factors	
Autonomy	Each learner can choose whether to participate or not, which and how many transparencies to pick, and where to position them
Competence	The game founds on motor skills and temporal reasoning skills
Relatedness needs	Each learner is stimulated to compete with others (the winner of the game is the kid that provide the composition most coherent with the temporal relations of the sentence)

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Table 8 The WHILE game

WHILE game	
Investigation objective	
To gather information about the mental representation of the WHILE relation	
Game material	
A4 paper sheets, colored crayons, voting paddles	
Game description	
Preliminaries	The investigator read a story and provides each child with a kit of crayons and an A4 paper
Goal	The goal of the challenge is to visually represent the sentence read by the investigator (containing a WHILE relation)
Moves	Each learner has to choose items from the game material to draw a scene
Engagement factors	
Autonomy	Each learner can choose whether to participate or not, which crayons to pick, and how to draw the scene
Competence	The game founds on drawing skills and temporal reasoning skills
Relatedness needs	Each learner is stimulated to compete with others (the winner is choose collectively by participating children who express their preferences by voting paddles)

3.3 Games' administration

While activities related to the design of the investigator kit correspond to the planning stage of the method (see Fig. 5), the running stage took place during ACTION-TAKING$_{PS}$ of the 1st and the 2nd iteration, involving, overall, fifteen sessions including about 400 children in the 7–9 age range.

Each game session was conducted according to the structure in Fig. 6 and involved three investigators: one action researcher, one expert of data gathering, and the class teacher. First of all, in each session the class teacher introduced the other investigators, explained the aim of the session, and established a playful atmosphere (*nurturing phase*). It was made clear that the participation in games was free and investigators explained to children the importance of the game session to ensure a sense of responsibility, essential to get reliable data (*motivational phase*). As to the *body phase*, it has to be observed that its implementation protocol was refined throughout our CAR process: since during the 1st iteration, when games were administered according to a predefined order, in several classes it was not possible to carry out all designed games within one hour because of children interruptions and unexpected events, the protocol was modified by adding Property IV in Table 4 and, during the 2nd iteration, re-designed games were administered according to a flexible plan taking into account: estimated duration of games, remaining time, topic coverage/priorities, warm-up/peak/relaxing cognitive curve, number of children. Not all games could be administered within a session and investigators used also knowledge on the past sessions to shape the current session so to have all topics covered throughout the whole process.

For each administered game, investigators introduced goals, moves, and rewards and provoked excitement (*energizing*). At the end of a game, they declared winners and delivered prizes (e.g., stickers) (*rewarding*). Investigators observed how children carried out game activities, while stimulating them in maintaining interest and supporting their requests, taking notes of each behavior interesting for later analysis, and taking photos of the game areas. For ethical and motivational reasons, at the end of the game session investigators made sure that

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each child got a reward (*closing phase*). Data gathered through games were then analyzed and used to populate the database previously designed (*reporting stage*).

In particular, as to games aimed at gaining knowledge on children mental representation of temporal connectives (see Tables 7 and 8), both for the BEFORE-AFTER game and the WHILE game after reading the whole story investigators read some individual sentences narrating situations of the story. Each sentence contained two events connected by a temporal relation, with the same character sometimes appearing in both events. Afterwards, children were challenged to produce *their representation of the sentence*: in the case of BEFORE-AFTER the goal of the challenge was to provide a composition of transparencies, positioned on a sheet of A4 paper, choosing transparencies depicting story characters from a kit previously received, while in the case of WHILE the goal of the challenge was to draw a scene using colored crayons freely chosen from a kit.

As to the BEFORE-AFTER game, the sentences read by the investigators are listed in Table 9. One may notice that the sentences differ for the position of the two events within the sentence relative to their chronological order: sometimes, the order within the sentence coincides with the chronological order, sometimes it does not. This was done to eliminate possible biases, and to investigate whether and how children grasp the distinction between the two orders, and whether age has influence on it. Given the short duration of individual challenges, it was possible to read more than one sentence within the same session.

As to the WHILE game, the sentences read by the investigators are listed in Table 10. One may notice that sentences *W1* and *W2* actually represent the same situation but differ for the position of the two events within the sentence. Also in this case the objectives were to eliminate possible biases and to investigate whether children grasp simultaneity of the two events and whether the age has influences on their understanding of it. Given the long duration of individual challenges, in this case the sentences were read in different sessions.

3.4 Analysis of results of time-related games

The analysis conducted during the EVALUATION/LEARNING$_R$ and EVALUATION/LEARNING$_{PS}$ stages of the 2rd iteration confirmed that quality and quantity of gathered data were adequate for the objectives of the project: quantity was considered sufficient for statistical significance and reliability was supported by evidences from teachers and parents of the involved children (gathered via contextual inquiries). The focus of the 3rd iteration was hence shifted from data gathering to data analysis, with the aim of defining users classes and personas, and exploiting the results of the overall set of games related to the different topics investigated. Here we focus only on the analysis of time-related games, which is the final goal of this paper.

Table 9 Sentences used for the BEFORE-AFTER game

Sentences for the BEFORE-AFTER game	
BA1	Gino the Chick picked some strawberries, then met Marta the Duck
BA2	After Gino the Chick met Marta the Duck, Tonino the Mouse told Gino “Stop here!”
BA3	Gino the Chick met Tonino the Mouse, then he went to the woods
BA4	Gino the Chick met the wolf after being arrived in the woods

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Table 10 Sentences used for the WHILE game

Sentences for the WHILE game	
W1	Mammy Hen and the others animals go to the wolf's house while Gino the Chick is telling stories to the wolf
W2	While Gino the Chick is telling stories to the wolf, Mammy Hen and the others animals go to the wolf's house
W3	While Mammy Hen is worried about her Little Gino, Gino the Chick is telling stories to the wolf

In particular, from the analysis of children-produced material collected during BEFORE-AFTER and WHILE games we could draw a number of considerations related to conceptualization of events, roles of event characters, (evolving) children mental models for the BEFORE-AFTER and the WHILE relations, hereinafter discussed. For this analysis we could benefit also from preliminary pre-project experiments [17]; the results have also been totally confirmed by a post-project experiment that we recently conducted with about 40 children from two classes, in the 6–9 age range.

Central role of characters Children may sometimes ignore some of the sentence/story elements (e.g., background elements or accessory objects) but for characters, which are always all represented in the drawings or selected from the kit of transparencies. Furthermore, as soon as writing skills are acquired, there is a recurring natural recourse to dual coding: characters are textually labeled by their names in children drawings (see Fig. 7a).

Cognitive load Children sometimes represent only one event of the read sentence, instead of the two events presented, hence showing difficulties in remembering the whole structure of the sentence.

Encapsulating events Children drawings show the recurring presence of a dividing item aimed at separating story elements belonging to different events, often encapsulating story

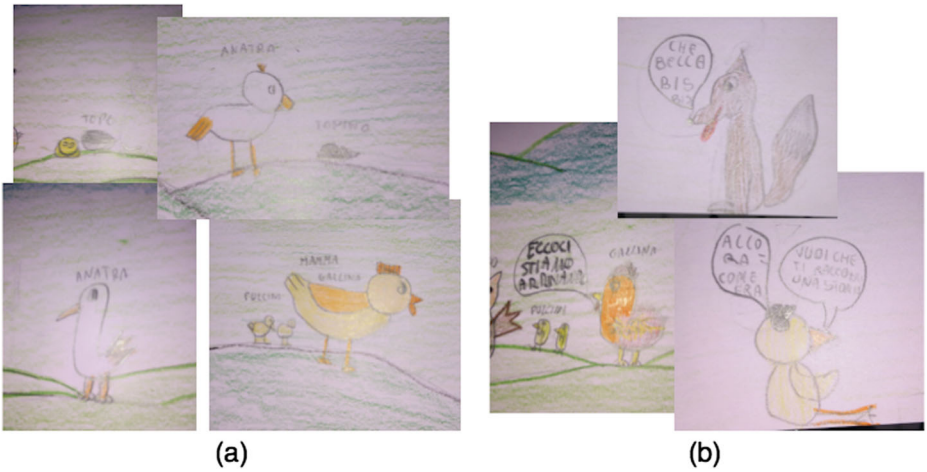

Fig. 7 *Dual coding from children* – **a**: characters are labeled with their names; **b**: visual elements are enriched with short sentences describing particular aspects of the events

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elements related to a single atomic event. Such dividers have sometimes an abstract nature (like, e.g., lines, bounding boxes, and free-hand lines, as in Fig. 8a and b), or otherwise a more concrete nature, based on consolidated children know-how (like, e.g., houses, fences, trees, and woods, as in Fig. 8c and d). It is worthwhile underlining that, when present, such concrete dividing elements do not belong to the sentence/story to be represented, but are introduced by children on an individual, free, and spontaneous basis. It is particularly interesting observing that when the sentence/story includes an element that could naturally be an "encapsulator" (e.g., a house), some children freely decide to use an additional abstract/concrete encapsulator inside it (e.g., a bounding box, a window, and a room) to border the event (as in Fig. 8e and f). This "event encapsulation" seems to be a primitive concept as witnessed by arrangement strategies adopted by first class learners in the BEFORE-AFTER game based on the transparency kit, where children often represent separately the two events; this is particularly evident when the same character is present in two distinct events of the sentence, and children decide to pick two instances of it from the transparency kit and to use them in the two groups of transparencies corresponding to the two events (see Fig. 9). Furthermore, again, as we also

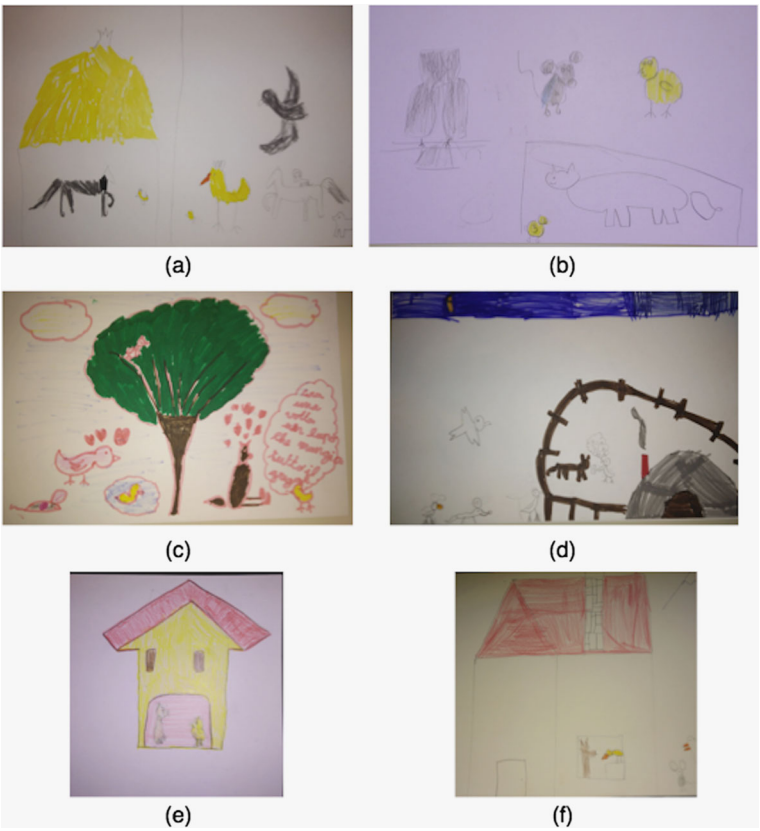

Fig. 8 *Encapsulating events* – Some drawings associated to the sentence "*Mammy Hen and the others animals go to the wolf's house while Gino the Chick is telling stories to the wolf*", illustrating the recurring presence of dividing elements. In particular, (1) the drawings in (**a**) and (**b**) show the use of abstract elements (respectively a vertical line and a bounding box), (2) the drawings in (**c**) and (**d**) show the use of concrete elements (respectively a tree and a fence), while (3) the drawings in (**e**) and (**f**) illustrate the case of additional encapsulation of events (respectively by an abstract element (bounding box) within the house, and a concrete element (a window))

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observed for characters, as soon as writing skills are acquired there is a frequent natural recourse to dual coding: visual elements are enriched with short sentences describing particular aspects of the event (see Fig. 7b). Also in this case, such textual elements do not belong to the original sentence/story.

BEFORE-AFTER *visualization* From the analysis of transparencies arrangements, it emerges the lack of any predominant spatial arrangement strategy, and, consequently, of any already formed precise mental model. We anyhow observe that, as age increases, sentence elements tend to be more frequently correctly positioned from left to right, coherently with their temporal relation rather than their position in the sentence. Hence, in order to favor the construction of a mental model coherent with customary adult-oriented timeline-based visual models, the TERENCE team decided that the TERENCE visualization of BEFORE-AFTER temporal relations has to be based on a left-to-right arrangement of related events.

WHILE *visualization* From the analysis of drawings, the combination of two aspects emerges, namely, spatial arrangement strategies and "reinforcing" drawing elements. As for *drawing elements*, we observe the recurrent presence of a common background (e.g., sky, woods, and grass) for events related by a WHILE relation and/or of an element linking such events (e.g., a path). As in the encapsulation of events, in both cases such elements do not belong to the sentence/story to be represented, but are introduced by children on an individual, free, and spontaneous basis (see Fig. 10).

As for *spatial arrangement*, three recurrent strategies to couple events in a WHILE relation may be observed: horizontal arrangement (where events are positioned according to the order in which they appears in the sentence), vertical arrangement, and diagonal arrangement (see Fig. 11); it is worth notice that diagonal and vertical arrangements are almost never encountered in children of the first class, while they start to appear in all the second classes of the experiments, together with the horizontal one.

Again, this fact testifies the lack of an already formed precise mental model, but also that the mental model tends to naturally evolve towards the vertical arrangement. In order to favor the construction of a mental model coherent with customary adult-oriented timeline-based visual models, the TERENCE team hence decided that the TERENCE visualization of

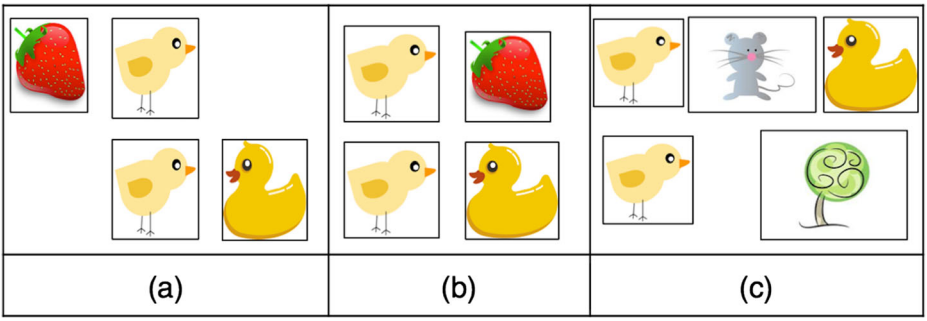

Fig. 9 *Again on encapsulating events* – When the same character appears in two events children pick two instances of the corresponding transparency to separately represent the two events. Transparency arrangements in (**a**) and (**b**) were associated to the sentence "*Gino the Chick picked some strawberries, then met Marta the Duck*", while the arrangement in (**c**) was associated to the sentence "*Gino the Chick met Tonino the Mouse, then he went to the woods*"(pictures (**a**), (**b**), and (**c**) reproduce the arrangements of actual transparencies, not shown here because of copyright reasons)

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Fig. 10 WHILE *visualization* – Some drawings associated to the sentence "*While Mammy Hen is worried about her Little Gino, Gino is telling stories to the wolf*", illustrating the recurring presence of linking elements. In particular, the drawing (**a**) shows a concrete element (a path), while the drawing (**b**) shows an abstract element (an L-shaped form). On the drawing (**a**) one may also notice the presence of a common background between the two events (the starred sky). Again, in both cases we underline the encapsulation of the events, either by concrete elements (the house and the wood in the drawing (**a**)) or by abstract elements (the bounding box in the drawing (**b**))

WHILE temporal relations has to be based on a vertical arrangements of related events (like in Allen's visualization), while maintaining some form of concrete linking element to offer children a representation including some familiar concept.

4 From requirements to the visual interface

Starting from results of the analysis on the children mental model of time, we are now able to discuss how they have been used to design the "read-n-play" visual environment through which TERENCE support the mastering of temporal connectives and overcoming the inadequacy of adult-oriented techniques singled out in Section 1.2. The main idea behind TERENCE is that the stimulation by the system integrates with the traditional stimulation by teachers. According to the advice of the experts involved in the project, the psycho-pedagogical stimulation plan in TERENCE is inspired by a traditional teaching strategy including reading the story and then analyzing the text via inference-making question answering, see for details [22]. Learning sessions mirrors a

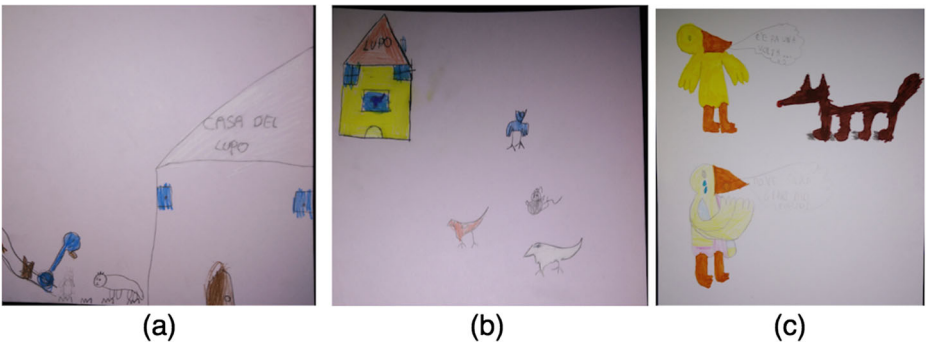

Fig. 11 *Spatial arrangement* – Drawings illustrating the three spatial strategies: horizontal in (**a**), diagonal in (**b**), and vertical in (**c**)

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warm-up, peak, and relaxing phases structure, specifically: (1) reading a story, silently – warm-up, (2) resolving related smart games for analyzing the story – peak, and (3) playing with other games able to relax the learners according to a their performances in the previous step – relaxing.

4.1 General aspects and reading activities

The interaction environment hence builds on a conceptual model based on the metaphor of the school and its elements (e.g., building, desk, book, classmates, and games). All displays are based on a simple common template including system communication on the left and interaction with content in the main area, as depicted in Fig. 12. In particular, Fig. 12a depicts the login display, while Fig. 12b shows the conversational nature of the system communication, carried out through an avatar. An interesting observation from our findings is related to the relationship that children engage with avatars: differently from adult-oriented games, where avatars are considered mostly as a personification of the player, in the children case the avatar was considered as a person distinct from the child, a kind of "helping character" acting as a surrogate of the teacher (usability evaluation, in fact, showed a high level of stress when children felt to be left alone with the system). For this reason we assigned to the avatar a conversational behavior, providing instructions, suggestions, and rewards in case of correct answers. Children can choose the avatar they prefer. As already mentioned in the Introduction, the stimulation plan is based on a *graded text simplification* approach: learners choose books and stories within books, and the system automatically provide the story version adequate to their current reading comprehension skill [3, 20]. In particular, once the child has chosen the avatar and a book to interact with (Fig. 13a), the system displays a map corresponding to the view of the book as a navigation space, including a path with locked and unlocked stories (a story gets unlocked when the learner passes the smart games associated with the previous story in the path) with no reference whatsoever to the reading comprehension levels, which are transparent to the child (Fig. 13b).

Central roles of characters Given the considerations about the central role of characters and the natural *dual coding* that children associate to them (see Section 3.4 and Fig. 7), the first activity that learners can do after choosing a story is browsing through "characters' cards", associating pictures of the story characters with a brief textual description of them (Fig. 14a). Whenever s/he wants, the learner can decide to abandon the character display to go the story.

(a) (b)

Fig. 12 Two screenshots based on the common template including system communication and interaction with content: **a** the login display, asking for username and password; **b** a message from the avatar ("*The story is over. Let's play with what you read in the story*")

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(a) (b)

Fig. 13 *Reading activities:* **a** choosing a book (the current selection is the book titled "*A summer to remember*"); **b** choosing a story (the picture shows the story titles from the "*A summer to remember*" book)

Granularity of the story structure It is worth noting that within the context of reading activities the granularity of the visual narration structure is that of episodes: the visual model addresses relative time occurrence of episodes while leaving it to the verbal narration to convey temporal information about events contained in episodes. As illustrated in Fig. 14b each story is structured as a sequence of illustrated episodes presented according to a focus + context carousel pattern, which proves efficient for exploring small sets, and hence well serves the structural characteristics of TERENCE stories, based on the model introduced in [64]: (1) stories are relatively "light", with a limited number of episodes, events, and characters; (2) episodes are temporally ordered, and (3) episodes are short.

Cognitive load The carousel pattern helps preventing memory problems due to high cognitive load by allowing children to focus on single episodes while maintaining a *global vision on the whole story* and on the order of episodes and freely browsing through them.

4.2 Events and smart games

The choice of not visually addressing individual events of episodes, in contrast to the literature about adult-oriented visual representation of temporal data and temporal connectives based on (crowded) timelines and visualization of intervals, is coherent

(a) (b)

Fig. 14 *Reading activities:* **a** browsing through characters (the current selection is "*Clare*", a "*40 years old woman, loving puppies*"); **b** browsing through story episodes (the episode in focus goes as "*While Ida was sitting in her tent reading a brilliant book ..*")

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with consolidated pedagogical approaches built on question-based games. For these reasons, in TERENCE individual visualization of events and of their temporal relations are associated to smart games and playing activities. We recall that since the aim of the stimulation is to drive children towards a mature mental model of time, our challenge is to build on concepts already possessed by children while at the same time retaining some basic features of adult-oriented temporal visualization to favor the transition.

Temporal games are classified into different level of difficulty to stimulate text comprehension according to increasingly demanding tasks:

- "BEFORE/AFTER *games*", including BEFORE, AFTER, and BEFORE-AFTER games, where the child has to reason about purely sequential events;
- "WHILE *games*", including BEFORE-WHILE, WHILE-AFTER, and BEFORE-WHILE-AFTER, where the child has to reason about both sequentiality and simultaneity.

All games are based on the same general pattern, illustrated in Fig. 15a: the content area is divided into three portions: a lower bar displays three tiles corresponding to three/four possible choices (depending on the game), a middle area displays the question to be answered, and an upper part depends on the specific games. In all cases the interaction is based on "drag-and-drop": the child has to select the tile corresponding to the correct answer and drag it into the specific element of the upper part. In temporal games the tiles represent events.

Encapsulation of events and granularity We observe that, though events have a duration, and in principle might hence be associated to time intervals, the level of granularity and the degree of indeterminacy of temporal information in learner-oriented stories make interval-based visualizations (such as techniques based on Allen's relations) not adequate. In TERENCE we adopted a simple dual-coded tile-based representation for the events, including an illustration and a verbal sentence, directly corresponding to the natural *encapsulation of events* singled out by the study in the children (see Fig. 15b). The illustration and the sentence recall what children read in the story without necessarily coinciding with pictures and sentences of the story, to stimulate inferences (on the other hand, the analysis of game results showed that children naturally rely on additional text).

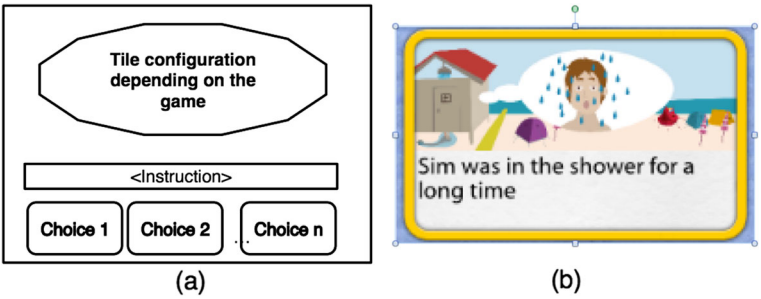

Fig. 15 *Playing activities:* **a** the general template; **b** an event tile

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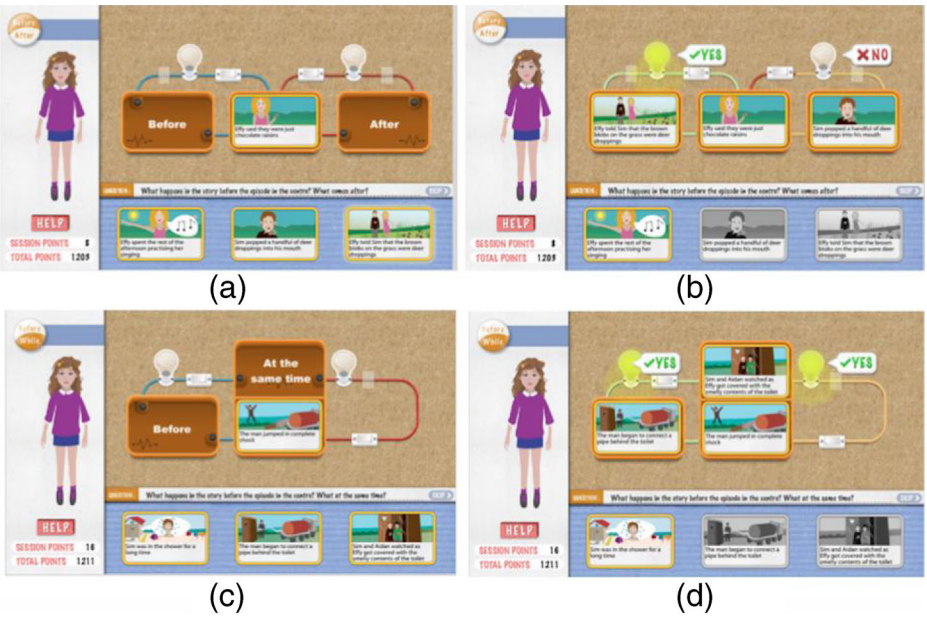

Fig. 16 *Temporal games*. Pictures (**a**) and (**b**) illustrate an example of BEFORE-AFTER game (the question and the try); in particular, in (**a**) the system proposes an event of the story in the middle tile of the upper part of the display ("*Effy said they were chocolate raisin*.") and asks a question ("*What happens in the story before the event in the centre? What comes after?*"; the learner has to pick one of the three options in the lower bar ("*Effy spent the rest of the afternoon practising her singing*", "*Sim popped a handful of deer dropppings into his mouth*", "*Effy told Sim that the brown blobs on the grass were deer droppings*"); in (**b**), after the learner has proposed an answer, the system signals its correctness. Pictures (**c**) and (**d**) illustrate an example of BEFORE-WHILE game (the question and the try); in particular, in (**c**) the system proposes an event of the story in the middle tile of the upper part of the display ("*The man jumped in complete shock*") and asks a question ("*What happens in the story before the event in the centre? What comes at the same time?*"; the learner has to pick one of the three options in the lower bar ("*Sim was in the shower for a long time*", "*The man began to connect a pipe behind the toilet*", "*Sim and Aidan watched as Effy got covered with the smelly contents of the toilet*"); in (**d**), after the learner has proposed an answer, the system signals its correctness

Reasoning about sequentiality (BEFORE/AFTER *games***)** The first row of Fig. 16 shows an example of BEFORE-AFTER game (the question and the try). The tile configuration for BEFORE game and AFTER game is clearly a simplification of this general case. It is possible to notice that we maintain the sequentiality of Allen's relationship while ignoring the distinction between 'before' and 'meets' cases (first two cases of Fig. 2). Concrete and intuitive visual metaphors (on/off light bulbs) are used to signal the correctness of the answer.

Reasoning about simultaneity (WHILE *games***)** The second row of Fig. 16 shows an example of BEFORE-WHILE game (the question and the try). The tile configuration for WHILE-AFTER and BEFORE-WHILE-AFTER games is defined coherently. It is possible to notice that we maintain Allen's suggestion to use parallelisms while ignoring the distinction among different cases of partial and complete overlapping ('overlaps', 'starts', 'during', 'finishes', and 'equals' in Fig. 2). Concrete and intuitive visual metaphors (on/off light bulbs) are used to signal the correctness of the answer. We also notice that the wires play the role of "reinforcing" linking elements, coherently with the results of the analysis of children drawings.

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5 Evaluation outcomes

Evaluating an educational system like TERENCE is a complex task that requires the assessment of both *system usability* and *psycho-pedagogical effectiveness*: if students cannot use the learning platform, they certainly will not learn through its use [8]. Within its framework of iterative design (see Fig. 1), the TERENCE evaluation was articulated in two expert based rounds and two user-based rounds.

According to the principle of repeated learning experience, evaluation rounds required a number of different sessions with the same children. Table 11 provides a synoptic view of the evaluations: the first column specifies the characteristics of the evaluation (expert-based/user-based, formative/summative, qualitative/quantitative), the second and third columns specify the issues and the release under evaluation, respectively, and finally the last column indicates the number of users involved. While an in-depth discussion on the overall TERENCE evaluation is beyond the scope of this paper (the interested reader may refer to [14]), here we focus on those aspects specifically related to the interaction with TERENCE temporal games and its impact on children learning (browsing and playing with temporal games was only one task out of eight tasks carried out by children during the usability evaluation).

5.1 Evaluating system usability

For evaluating the system usability, customary methods were used [56]: heuristics evaluation and cognitive walkthrough for the expert-based evaluation and direct observation and controlled experiment for the user-based evaluations. During the two user-based evaluation rounds, *qualitative data* were gathered both via direct

Table 11 The TERENCE project evaluation rounds

Evaluation Characteristics	Issues	Release (Month)	Involved Users
1st expert-based Formative Qualitative	Usability Curricular material (e.g., stories)	Prototypes (March 2012)	About 10 domain experts of text comprehension and interaction design
1st user-based Formative Qualitative	Usability Learning outcomes	1st release (June 2012)	About 170 learners (deaf and hearing)
2nd expert-based Formative Qualitative	Smart Games revision and production	2nd release (Sept 2012)	About10 domain expert of pedagogy
2nd user-based Summative Quantitative & Quantitative	Usability Learning outcomes	3rd release (March 2013)	About 830 learners (deaf and hearing)

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observations (e.g., facial expressions and by tracking comments using a structured schema) and via inquiry (indirect questions to children) at critical points (e.g., if the child explicitly asked for help, if the child seemed lost, if the child seemed upset). During the 2nd user-based evaluation, when the last prototype went live, the following *quantitative data* were gathered through log files:

- The story titles chosen by learners;
- For the reading task, the start and end time for reading the selected book;
- For each game instance, the time for their resolution before the game was over.

In general, evaluation aimed to investigate (i) if the issues carried out during an evaluation at stage i, were satisfactorily fixed in the TERENCE release at the evaluation stage $i+1$; (ii) the interaction quality; (iii) the satisfaction of the learners in using the system and the interest in using it in a non-school context. Table 12 report the main usability evidences emerged when children played with temporal games, grouped in disjoint sub-categories: choices, interaction modalities, feedback, user satisfaction, and time.

5.2 Evaluating psycho-pedagogical effectiveness

For evaluating the psycho-pedagogical effectiveness, a variety of commonly used quantitative and qualitative techniques were utilized [29]. In particular, evaluation aimed to investigate: (i) pre/post difference in an experimental group and in the single schools that made up of the experimental group; (ii) pre/post difference of the experimental group and the single schools with respect to a control group, and (iii) whether a different effect can be identified in poor comprehenders than in good comprehenders.

As summarized in Table 13, the evaluation showed that (1) the stimulation plan significantly improved comprehension in the experimental group, (2) TERENCE improved reading

Table 12 Evidences singled out during the usability evaluation of temporal games

Sub-categories	Evidences about temporal games
Choices	(a) Children easily used captions, they resulted usable, no problems emerged
	(b) Children had no problems in managing temporal game choices
	(c) Children had no problems in understanding if the choices were available or not
	(d) Children appreciated the grey-out effect for highlighting the unavailable choices
Interaction Modalities	(a) Children had no problems in using the drag and drop
	(b) Children appreciated the affordance of all games
Feedback	(a) When the solution feedback appears, children noted it, and were interested in understanding what was the correct solution.
	(c) Consistency and explanatory feedback resulted in general clear
User Satisfaction	(a) Instruction resulted in general clear, thanks to the system tutorial .
	(d) Children appreciated the visual metaphors of temporal games.
	(e) Children liked the visual metaphors of the feedback.
	(f) Children asked to play TERENCE at home, to finish reading “their” book.
Time	(a) Times for playing decreased using the system.
	(g) Learners’ precision in resolving time games increased using the system.

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Table 13 Evidences singled out during the evaluation of psycho-pedagogical effectiveness

Aims	Group	PRE	POST	P values and notes from statistical analysis
(i)	Experimental	**9,01 %** of learners appeared poor comprehenders	**2,50 %** of learners appeared poor comprehenders	By comparing the pre/post effect only in the experimental group, we have $P = 0.0234$: this means that at the end of the experiment, we have a statistically significant reduction in the number of poor-comprehenders
(ii)				By comparing the pre/post effect in the experimental wrt the control group, $P < 0.0001$; this means that the TERENCE improved reading comprehension (the control group should have removed all other possible effects, e.g., natural cognitive growth)
(iii)	Good comprehenders	**8.67** is the average MT comprehension score	**8.73** is the average MT* comprehension score	By comparing the improvements in reading comprehension in both groups, $P = 0.3665$; this means that the TERENCE stimulation plan improves comprehension both in poor and good comprehenders (*$0 < MT < 10$)
	Poor comprehenders	**5.31** is the average MT comprehension score	**5.88** is the average MT* comprehension score	

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comprehension also in comparison with a control group, and (3) TERENCE improved comprehension both in poor and good comprehenders, demonstrating, de facto, that the choices implemented in the visual interface in terms of visual representations for temporal relations are appropriated for a children-oriented system. Moreover, the absence of dropout confirmed the effectiveness of the classical psycho-pedagogical stimulation plan.

6 Conclusions

In this paper we discussed the challenges we faced to design/implement an adaptive learning system supporting 7–11 years old "poor text comprehenders". Specific requirements emerged, like (*R1*) the necessity of designing in parallel the mutually dependent psycho-pedagogical stimulation plan and the system modules realizing it, (*R2*) the definition of a psycho-pedagogical assessment strategy, (*R3*) the necessity of large-scale studies within users' inquiry, assessment, and evaluation activities to ensure statistical significance of results, and (*R4*) the necessity of defining suitable children-oriented visual representations for temporal structures of stories. Existing consolidated techniques turned out to be inadequate because not compliant either with the age of the target users or with ethical and organizational constraints coming from the specific context of use (schools).

To cope with the requirements *R1* and *R2*, we specialized the User Centered Design Schema to meet TEL requirements, in order to emphasize the role of the psycho-pedagogical stimulation plan and related issues. To cope with requirement *R3* we launched an AR project, within the framework of the analysis of the context of use of the 1st iteration of the TEL-oriented UCD, which allowed us to define, validate, and apply to the problem at hand a novel children-oriented game-based data gathering technique, while at the same time translating the experience into scholarly knowledge related to design and evaluation of children-oriented applications. To cope with requirement *R4* we exploited the results of the analysis of the context of use relative to the children mental model of time to design a children-oriented integrated "read-n-play" visual environment able to force the acquisition of the capability of reasoning about temporal events within a story. While for pedagogical aspects of the stimulation plan we refer to [3, 20] and for architectural/implementation aspects we refer to [19] in this paper we focused our attention on aspects related to the design of the visual aspects of the system.

In particular, the novel data gathering technique allowed us to comply to our overall requirement asking us to gather *high quantity of high quality data from many users in short sessions with medium size users' groups*. Coherently with the principles of Action Research, the technique was defined, revised, and validated "on the field" within the framework of an iterative process continuously involving psycho-pedagogical experts and teachers, who cooperated with the action researchers to assess the technique with reference to game design, experience with the class, and quality of gathered data. As to *game design,* it was observed that the multidisciplinarity of the team allowed us to naturally derive children-appropriate games. As to the *experience with the class,* it was observed that there was a smooth integration between teachers and researchers as well as between game-based data gathering activities and regular school activities (wrt both contents and modus operandi); teachers remarked the value of the activities for children; the flexible game administration plan succeeded; furthermore deans appreciated the unobtrusiveness of the approach with respect to school resource involvement, cost, and needed time. As to *gathered data*, their reliability was supported by

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Table 14 Comparison with existing techniques

Adults/Children	Techniques	Similarities	Plus
Adult-oriented and adaptable to children	Survey	High quantity of data statistically analyzable	High data quality
	Interviews	Direct interaction with users	High quantity of data in relatively short time
	Questionnaires	High quantity of data in relatively short sessions	High data quality
	User observation and field studies	Viewing users in their real context	No interference with routine activities but integration of "new" activities in the routine
	Brainstorming	High data quality	High data quantity
Children-oriented	User Interface Drawing	High quantity of data in relatively short sessions	High data quality
	Idea Maps	High quantity of data in relatively short sessions	High data quality
	Kid Reporters/Networking news	High quantity of data in relatively short sessions	No interference with routine activities but integration of "new" activities in the routine
	Mission from Mars	Playful and motivating framework, and high data quality	No interference with routine activities but integration of "new" activities in the routine

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evidence from teachers and parents of the involved children (gathered via contextual inquiries); furthermore, despite withdrawal from few children, since children were adequately stimulated the participation was high and consequently we got a high rate of returned data. Table 14 reports also a synoptic view of a comparison with existing techniques, showing how we were able to preserve their positive aspects while overcoming their limitations.

A set of games was then designed according to the technique guidelines, in order to investigate the topics identified by the preliminary study. In particular, the analysis of the material collected during time-related games allowed us to draw a number of considerations related to the children conceptualization of events, the roles of characters, and the evolving children mental models for the BEFORE-AFTER and the WHILE relations, which were the basis for the design of the TERENCE "read-n-play" visual environment. There emerged, in particular, a natural recourse to dual-coding, both for characters and events, a strong necessity of event encapsulation, and, as the age increases, a trend towards conventions of adult-oriented timelines for what concerns sequentiality and simultaneity of events. The visual environment was hence designed based on these considerations, paying attention to the degree of determinacy/granularity of temporal information, so to define visualization structures able to drive children towards a mature mental model of time: the visual model builds on concepts already possessed by children while at the same time retaining some basic features of adult-oriented temporal visualization to favor the transition.

As to the evaluation, the analysis of data gathered in a pre-post test involving two groups of users (experimental and control groups) during the summative evaluation showed that (1) the stimulation plan significantly improved comprehension in the experimental group, (2) TERENCE improved reading comprehension also in comparison with a control group, and (3) TERENCE improved comprehension both in poor and good comprehenders, demonstrating, de facto, that the choices implemented in the visual interface in terms of visual representations for temporal relations are appropriated for a children-oriented system. Furthermore, beside the pedagogical effectiveness in terms of improvement in text comprehension, clearly emerged the positive attitude of children towards a "gamified" environment, with children engaging competition with each other, showing eagerness to use the system, and developing more and more affection towards their avatars and books.

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